

Environmental impact assessment of DGFIP Gen AI Platform

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In the wake of the rapid integration of Generative AI within French public administrations, the **Direction Générale des Finances Publiques (DGFiP)** has launched its GenAI platform. While these models offer transformative potential for administrative efficiency, their environmental footprint remains a critical concern for public accountability.

The objective of this report is to provide a **Life Cycle Assessment (LCA)** of the DGFiP's AI infrastructure. We go beyond carbon emissions from inference by evaluating every stage of the AI value chain: from the raw materials required for GPU manufacturing to the operational energy of daily inferences and the final management of hardware waste.

We also created a custom **Energy Attribution Tool** developed to resolve the granularity gap in data center monitoring. While standard PDU meters only provide aggregate consumption, our tool uses a multi-factor weighting logic incorporating numerical precision (quantization), active parameter counts (MoE), and GPU idle power baselines to isolate the carbon footprint of each specific model. This could help the DGFiP with model selection based on CO_2 emissions per request.

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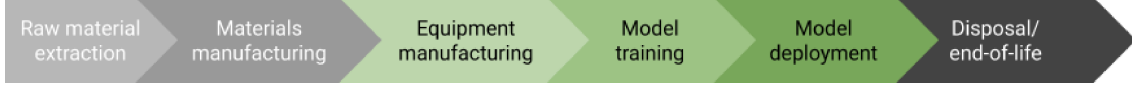
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General framework

1.1 Environmental Life-Cycle Assessment (LCA)

To evaluate the environmental footprint of DGFIP platform, we conducted a literature review covering both direct and indirect emissions across all stages of the AI lifecycle, following the methodologies established in recent research (Berthelot et al. 2024; Plociennik et al. 2025; Ciambrone 2018). We focused on CO_2 emissions but other outputs can be taken into account such as material consumption (including rare earth elements), and water consumption (Luccioni, Viguiet, and Ligozat 2023).



1.2 Distribution of emissions across scopes

The environmental burden is not uniformly distributed. Based on the life-cycle analysis of large-scale models (Luccioni, Viguiet, and Ligozat 2023), we categorize emissions into three distinct segments:

- **On-site emissions (< 2%)**: It is the CO_2 emission from the data center. These include emissions such as the leakage of refrigerant gases in cooling systems and the combustion from diesel back-up generators.
- **Direct energy emissions (70%)**: This represents the dominant share, primarily driven by the electricity consumption of local servers during the inference phase when the platform is operational.
- **Embodied emissions (20% to 30%)**: These originate from the value chain, encompassing the manufacturing, transportation, and end-of-life processing of hardware. Crucially, this scope also includes a amortized fraction of the initial training debt, i.e. the environmental cost incurred during the model's creation (Luccioni, Viguiet, and Ligozat 2023).

These percentages are not static; they fluctuate significantly depending on the carbon intensity of the energy grid, the specific hardware architecture (e.g., TPUs vs. GPUs), and the parameter size of the models used.

This report is structured into four main phases, following the standard environmental impact assessment categories:

- **Phase I**: Hardware and infrastructure
- **Phase II**: Training
- **Phase III**: Inference
- **Phase IV**: End-of-Life

Phase I: Hardware and infrastructure

The first phase of our analysis focuses on the physical layer of the platform, encompassing the manufacturing of specialized AI hardware and the overhead of the data center infrastructure.

2.1 Description of the infrastructure and the models at DGFIP

The infrastructure is characterized by GPUs distributed across two environments:

- **Production environment:** 8 servers, each equipped with 4 GPUs, totaling **32 GPUs**.
- **Development environment:** 2 servers, each equipped with 4 GPUs, totaling **8 GPUs**.

All of this compute power relies on **NVIDIA H100 80GB HBM3** units, each featuring a peak power consumption of approximately **700W**. These are hosted in a facility with a measured **PUE (Power Usage Effectiveness) of 1.7**, meaning that for every watt consumed by the GPUs, an additional 0.7 watts is required for cooling and power distribution.

The different AI models used are allocated **24 H100 GPUs**.

Model name	Deployment configuration	Hardware
Llama-3.3-70B	1 instance	4 GPUs
Gpt-oss-120b	3 instances	6 GPUs (2 per inst.)
Qwen3-VL-32B-Instruct	2 instances	4 GPUs (2 per inst.)
Gpt-oss-20b	2 instances	2 GPUs (1 per inst.)
Qwen2-1.5B-instruct	4 instances	4 GPUs (1 per inst.)
Whisper-v3-Turbo	1 instance	1 GPU
gte-Qwen2-1.5B	1 instance	1 GPU

Table 1: Current DGFIP model distribution and GPU allocation

2.2 Environmental footprint of H100 chips

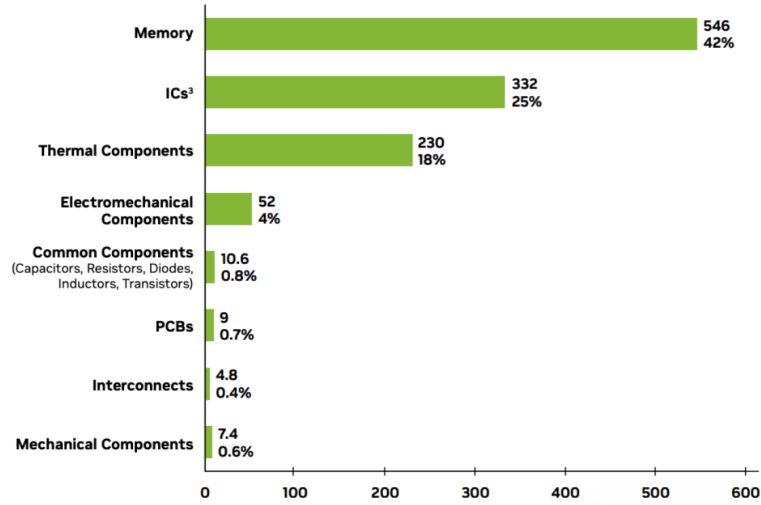
To quantify the impact of the computing power, we rely on the “Cradle-to-Gate” assessment of the NVIDIA HGX H100 (80GB HBM3) architecture, which covers the lifecycle from raw material extraction to the factory exit. A single unit in the fleet consists of a baseboard equipped with 8 GPUs. The carbon footprint breakdown for one unit is distributed as follows:

- **Components (91%):** Primarily driven by the high-intensity manufacturing of Memory and Integrated Circuits (ICs)
- **Assembly and Transport (9%):** Covering the final construction and global logistics.

According to these specifications, the baseline for one standard HGX baseboard is **1,312 kg CO₂e**. For the current DGFIP fleet, which operates 24 GPUs (3 baseboards), the total manufacturing footprint reaches **3,936 kg CO₂e (3.9 t)**. To put this into perspective, this is equivalent to **driving approximately 18,000 km** in a standard thermal car or the manufacturing of 15 full desktop computers (excluding monitors).

Beyond carbon metrics, the manufacturing process is significantly resource-intensive; a single advanced GPU can consume over 30,000 liters of water during production (ADEME carbon impact calculator).

HGX H100 PCF: Material Breakdown by Component Type (Kg CO₂e/Unit)



However, it is important to note that when amortized over a standard 4-year usage period, the manufacturing phase represents a relatively small portion of the total lifecycle emissions compared to the operational phase (Schneider et al. 2025).

2.3 Data center efficiency

The environmental impact of the hardware is amplified by the efficiency of the hosting environment, measured by the Power Usage Effectiveness (PUE). The PUE represents the ratio of total energy consumption to the energy delivered to the IT equipment.

Currently, the DGFIP data center operates at a PUE of 1.7, meaning for every 1 kWh consumed by the GPUs, 0.7 kWh is wasted on non-computing support (primarily cooling). Reducing this to a target of 1.4 is a priority for lowering the total energy demand.

To ensure a holistic view, our framework also identifies secondary emission sources often omitted from standard reports:

- **Refrigerant leakage:** The cooling systems often utilize gases such as R-134a, which has a Global Warming Potential (GWP) 1,430 times higher than CO_2 .
- **Back-up power:** Diesel generators contribute to emissions during mandatory monthly testing and during any power outages.

Strategic recommendation for DGFIP

- **Hardware longevity:** Extend the lifecycle of current GPUs as long as possible to maximize the **amortization** of manufacturing emissions over time.
- **Efficiency-driven chips:** When hardware replacement becomes inevitable, prioritize high-efficiency architectures. For instance, **NVIDIA B200** chips offer significantly superior energy-per-token performance compared to previous generations.
- **Advanced cooling solutions:** Transition from air cooling to **water cooling systems**. This shift can potentially reduce the PUE to ~ 1.1 , cutting the total operational footprint by approximately 30%.
- **Frugal infrastructure (Heat recovery):** Implement circular energy solutions by **repurposing waste heat** generated by servers to provide heating for nearby buildings.

Phase II: Initial training

3.1 General framework to understand DGFIP’s responsibility

Even if DGFIP did not train its own models internally, these are “upstream” inputs of the AI life-cycle. In this sense, these models are “pre-processed” resources that arrive with an inherited carbon debt due to their initial development. In a rigorous LCA, an institution accounts only for a portion of training emissions, amortised over the total expected inference volume for each downstream fine-tuned version.

However, direct methodologies for attributing this fractional responsibility remain underdeveloped. The primary hurdle is data asymmetry: without knowing the total global inference volume of a model, a single downstream user cannot mathematically determine their specific “slice” of the initial training debt. Schwartz et al. 2019 underline the lack of reporting standards and transparency from model creators which prevents downstream users from calculating precise efficiency metrics.

Therefore, a proxy for the volume of downstream could be the popularity of the model, where more popular models are seen to have more downstream uses and therefore the fractional responsibility of downstream users is seen to be much smaller. However, this logic creates a perverse incentive. Promoting “efficient” popular models can drive up total demand, eventually leading to the training of even larger, more energy-intensive successor models. For example, GPT-3 was very popular and proved the utility of LLMs. The massive influx of users then justified the transition to GPT-4 which has a ten times higher number of parameters (and thus energy usage). This is what Luccioni, Viguiet, and Ligozat 2022 consider as the Jevons’ Paradox, whereby improvements in efficiency (lower upstream carbon emissions per user in this case) may paradoxically spur increased consumption. Therefore, we cannot simply rely on the popularity of the model to drive our choice: more popular models do not necessarily induce less carbon debt.

Subsequently, we will not be able to assign “hard” CO₂e values to the initial training that could have been cleanly aggregated with DGFIP’s inference emissions. Any single final number for the platform’s emissions would be statistically unreliable and subject to significant margins of error.

Strategic recommendation for DGFIP

Rather than relying on imprecise fractional attributions, we propose that DGFIP evaluate and select models based on their absolute initial training emissions. By treating a model’s total inherited carbon debt as the primary benchmark, DGFIP can ensure that its selection criteria are directly proportional to the environmental cost of a model’s creation — even if its specific usage slice remains a fraction of that total.

3.2 Emissions of the initial training of the pre-trained models used by DGFIP

The models used by DGFIP are sourced from HuggingFace. While HuggingFace promotes transparency through its Carbon Emissions feature (*Displaying carbon emissions for your model* · Hugging Face 2026), the latter has not experienced wide-spread adoption so far. According to Castaño et al. 2023, 99% of models do not report emissions-related information, including foundational models like BERT and GPT-2. This lack of data obscures the environmental impact and hinders the adoption of carbon-efficient practices. However, the authors propose a five category classification scheme that can help us understand the initial training emissions of the models used by DGFIP.

3.2.1 Certified energy efficiency

This category requires optimised emissions and adherence to rigorous, established energy efficiency standards. None of the models utilised by DGFIP or available on HuggingFace fall into this category.

3.2.2 Energy awareness

This category designates models that report both carbon emissions and technical context (hardware, location of training, dataset size...). The Llama 3.3-70B used by DGFIP falls into this category. Indeed, the model card accessible on HuggingFace *meta-llama/Meta-Llama-3-70B-Instruct · Hugging Face* 2024 indicates training emissions of 11 390 tons CO₂e for the entire collection and 2 040 tons CO₂e for this specific model. They also include contextual information: the model was trained for 39.3M GPU hours on H100-80GB with power 700 TDP.

3.2.3 Basic emissions reporting

This category is for models which provide a raw CO₂e figure but lack the context to make the data fully actionable. None of the DGFIP models fall in this category.

3.2.4 Context reporting

This category designates models that do not provide a final emission number but report the technical variables required to estimate it. Four of the models that the DGFIP uses fall in this category.

For these, some tools like *Machine Learning CO2 Impact Calculator* 2026 or *Green Algorithms - Classic view* 2026 can enable rough approximations of the training emissions of the models.

The Qwen2.5-Coder-32B-Instruct model (Hui et al. 2024) used by DGFIP does not indicate training time but reported training on 5.5 trillion tokens (Kaplan et al. 2020). The hardware used by Qwen is Alibaba Cloud (typically H800 or A800 clusters), with PUE of 1.2 (Alibaba northern China data centers) and carbon intensity 550g/kWh since the Chinese grid is coal dependent. Thus, an estimation using the ML CO₂ gives 466.4 tons CO₂eq (*Alibaba Progresses Towards Carbon Neutrality Goals and Digital Inclusion* 2026).

The gpt-oss120b model card (OpenAI et al. 2025) indicates a total of 2.1 million H100 hours to complete training. OpenAI's training partner for GPT-OSS (Microsoft Azure) reports a global weighted PUE of 1.12 for its newest data centers. Carbon intensity in the US is around 300g/kWh. An estimation using the ML CO₂ impact is 1087 tons CO₂eq.

The gpt-oss20b model card indicates that it needed 10x fewer H100 hours to complete training compared to the 120b version. An estimation using the ML CO₂ impact is 108,7 tons CO₂eq.

However, these are approximations based on the predicted training locations, and using similar hardware conversions. These are not to be taken in absolute terms due to inherent measurement and data uncertainty.

3.2.5 Unknown emissions and context

The last category is the one in which most models fall. It is a category where models do not report any emissions-related information (neither emissions nor context) which clearly hides their

environmental impact and inhibits the development of carbon-efficient practices.

Mistral reports 20 400 tonnes for Large 2 (*Our contribution to a global environmental standard for AI* / Mistral AI 2026), but provides no specific data for Mistral-Small-24B-Instruct. (*mistralai/Mistral-Small-24B-Instruct-2501* · Hugging Face 2026)

Whisper-Large-v3 discloses that training took place for more than 5M hours, but no more detailed information is provided (*openai/whisper-large-v3* · Hugging Face 2022).

Qwen2-1.5B-Instruct reported being trained on a “large amount of data” with no specific token count or hardware hours (*Qwen/Qwen2-1.5B-Instruct* · Hugging Face 2026).

The e5-large model is said to be continually trained. They only mention 30k training steps and 32k batch sizes but this is not sufficient to calculate any carbon debt (*intfloat/multilingual-e5-large* · Hugging Face 2025).

When emissions and context are unknown, DGFIP must rely on proxy indicators. Especially that calculating precise energy consumption is inherently challenging due to the requirement to trust developer-reported data. Therefore, it is possible to use indicators such as training location, infrastructure types, dataset size of training, model size and parameter count.

a) Training location The geographic region of training is one of the most significant factors in a model’s operational emissions (Dodge et al. 2022). It is possible to refer to the IEA maps on carbon intensity in real time by region (*Real-Time Electricity Tracker – Data Tools* 2026). However, training locations for AI models are often closely guarded secrets due to the massive competitive advantage (and energy costs) involved. Still, corporate reports, data center expansion announcements, and environmental audits can reveal locations for the major tech companies and their models.

For reference, GPT trains most of its models in West Des Moines, Iowa. They detail the fact that their “supercomputer in a cornfield” was built specifically for OpenAI (Bach 2026). Newer clusters in Wisconsin and Atlanta are also cited in Microsoft’s 2025/2026 infrastructure blogs (Bolgar 2026).

Llama trains its models in Altoona, Iowa & Prineville, Oregon. They confirmed Llama 3 was trained on two 24k H100 GPU clusters (*Building Meta’s GenAI Infrastructure* 2024). Meta’s Sustainability Reports list Iowa and Oregon as their primary high-compute hubs.

Qwen trains its models in Zhangjiakou & Ulanqab, China. These regions are identified as the core of their intelligent computing infrastructure (*Zhangjiakou Data Centers - 18 Facilities from Operators* 2026).

Mistral trains in both the US & the EU. Indeed, Mistral’s partnership with Microsoft grants them Azure supercomputing access (Boyd 2024). Additionally, a CoreWeave Case Study explicitly states their newer models were trained “100% on CoreWeave infrastructure” (*Mistral AI Case Study* / CoreWeave 2026).

b) Infrastructure Types Power Usage Effectiveness (PUE) measures data center efficiency by comparing total facility energy to the energy used by the hardware. While the IEA reports a global average PUE of 1.58, modern AI facilities target a much leaner 1.10 to 1.15. This distinction is critical for DGFIP’s assessment because newer architectures like the NVIDIA A100 and H100 are significantly more power-hungry; without efficient cooling and power distribution (low PUE), their high power density can inflate total energy consumption by up to 50%.

For DGFIP, having transparency regarding PUE and hardware architecture is essential to accurately quantify the “inherited carbon debt” of pre-trained models. When these metrics are

disclosed in Model Cards, they allow for high-confidence auditing; when they are absent, DGFIP must assume a worst-case scenario based on global averages, potentially masking a high environmental cost. Tracking these infrastructure indicators ensures that the choice of model accounts for the efficiency of the “factory” where it was produced, not just its performance.

Regarding the models in your selection, transparency varies: Llama-3.3-70B is documented as being trained on high-efficiency clusters with a PUE of 1.15 (Meta, 2024). Conversely, models like Qwen2.5-Coder-32B, Mistral-Small-24B, and Whisper-v3-Turbo do not disclose the specific PUE of their training facilities, meaning DGFIP must assume a higher environmental risk based on the global average.

c) Training volume A model’s inherited carbon debt scales with the total tokens processed (dataset size multiplied by epochs); however, training often suffers from diminishing returns where the final 1% of accuracy disproportionately inflates energy costs. While early-stopping can significantly optimize this debt compared to a full life-cycle run, environmental inefficiency often arises from small batch sizes that fail to saturate high-power GPUs, leaving hardware idling while consuming near-peak power (Patterson et al. 2021). Within DGFIP’s portfolio, transparency is inconsistent: Llama-3.3-70B and Qwen2.5-Coder-32B provide clear data on their 15T and 5.5T token datasets (Meta, 2024; Alibaba, 2024), whereas Mistral-Small-24B and Whisper-v3-Turbo lack specific epoch and batch size disclosures.

d) Model size The model size, represented by parameter count, is the most consistent metric available for assessment and serves as a reliable predictor for the absolute floor of the training’s environmental cost. Research demonstrates a strong linear correlation between a model’s size and its footprint; specifically, a model ten times larger will typically generate impacts one order of magnitude greater than a smaller counterpart for the same volume of generated tokens (*Our contribution to a global environmental standard for AI* / Mistral AI 2026). For DGFIP, this means that even without precise data on the training duration, the parameter count allows for a reliable comparative audit of the embodied emissions inherited at the moment of deployment. This makes the size of models like Llama-3.3-70B and GPT-OSS-120B the primary driver of their high initial carbon footprint, regardless of how efficiently they are later utilised in inference.

3.3 DGFIP models table

Table 2: DGFIP pre-trained models: initial training assessment

Model Name	Model Size (Parameters)	Training Location (Grid Intensity)	Hardware & Infra (PUE)	Dataset & Config (Frugality)
GPT-OSS-120B	X (Very High - 117B)	~ (US Grid)	~ (Azure - 1.12)	~ (Standard MoE)
GPT-OSS-20B	~ (Moderate - 21B)	~ (US Grid)	~ (Azure - 1.12)	V (Efficient 4-bit)
Llama-3.3-70B	X (High - 70B)	~ (US - Mixed)	~ (Meta - 1.15)	X (15T Tokens)
Qwen2.5-Coder-32B	~ (Moderate - 32B)	X (High - China)	~ (Alibaba - 1.20)	~ (5.5T Tokens)
Mistral-Small-24B	~ (Moderate - 24B)	V (Low - France)	~ (Standard Cloud)	? (Unknown)
Whisper-v3-Turbo	V (Low - 809M)	~ (Mixed/US)	? (Unknown)	V (Pruned Layers)
gte-Qwen2-1.5B	V (Low - 1.5B)	X (High - China)	~ (Efficient PUE)	~ (High Volume)
dgfip-e5-large	V (Low - 335M)	~ (US Grid)	~ (Azure - 1.12)	V (Fine-tuned on 1B pairs)

Legend : V - favourable indicator, ~ - moderate / partial data; X - unfavourable indicator; ? - unknown.

Strategic recommendations for DGFIP

Training and development can represent between 50% and 65% of lifecycle impact for models in early deployment (Luccioni, Viguier, and Ligozat 2022; Morrison et al. 2025) and this can be reduced under 10% of total energy for models processing large volumes of queries over extended periods (Bose 2025).

Therefore, even though DGFIP uses pre-trained models, the training phase is non-negligible as it represents an inherited carbon debt that characterises the model's baseline.

To account for this, DGFIP should prioritise models with transparently reported emissions data. When such data is missing (which is often the case), the key indicators such as size, location and hardware can be reliable proxies. It is important to note that the goal is to choose a model with low initial training emissions, and fortunately, models with high training footprints are typically those with excessive parameter counts that also lead to higher energy consumption during inference.

By prioritising models with published Model Cards and carbon emissions or context reporting, DGFIP can effectively choose models to mitigate its inherited carbon debt of training while also leading to greater transparency across the AI supply chain.

Phase II Bis: Fine-Tuning

While DGFIP does not currently fine-tune any of its models, this may become relevant as its AI platform matures.

It is worth noting that fine-tuning is exponentially less compute-intensive than pre-training. For a 7B parameter model using efficient techniques like LoRA, the carbon impact can be as low as 50–100g CO₂e (equivalent to driving 250 to 400 meters in a standard petrol car, or the energy used to boil a few litres of water) (Ludvigsen 2024).

Should DGFIP engage in fine-tuning in the future, the environmental cost would remain negligible relative to the inherited training debt of the base model. The impact would however vary considerably depending on the fine-tuning method (LoRA vs. full fine-tuning), the model architecture (dense vs. sparse), the number of epochs, and the size of the training dataset.

Strategic Recommendation for DGFIP

Given the negligible carbon cost relative to pre-training, the fine-tuning phase should not be a decisive factor in model selection but adopting efficient methods remains good practice and ensures consistency with DGFIP's overall carbon reduction objectives.

Phase III: Inference

5.1 Development of the inference assessment tool

In response to the DGFIP’s requirements, we developed a dedicated assessment tool to quantify the carbon footprint of AI inference. The objective was to move beyond theoretical estimates and provide a data-driven methodology based on actual infrastructure logs.

The tool follows a three-step pipeline:

- **Inputs:** We collect energy data from Dell iDRAC (`extract_iDRAC.py`) and server activity logs (`parse_logs.py`) to track token processing.
- **Processing:** Data is aggregated using the `Method_IAG.xlsx` template, which calculates energy distribution across the development and production clusters.
- **Output:** The system generates a quarterly report tracking key performance indicators (KPIs) such as total kWh, CO_2e emissions, and overall efficiency.

5.2 Methodology and efficiency factors

The carbon footprint of inference is calculated using the following formula:

$$CO_2 \text{ emissions} = \text{Energy Consumption (W)} \times \text{PUE} \times \text{Electricity Mix} \quad (1)$$

However, the efficiency of inference depends heavily on how the model is squeezed and managed. Our methodology applies three critical corrections to attribute energy usage accurately:

- **Quantization:** The precision of the model significantly impacts compute requirements. For instance, FP8 models use approximately 50% less compute than their BF16 equivalents.
- **Architectural optimization (MoE):** Models using Mixture of Experts (MoE), such as *gptoss*, are more efficient as they only activate a small fraction of parameters per request.
- **Batch size optimization:** To ensure every watt produces the maximum number of tokens, we aim to maximize the batch size, processing multiple requests simultaneously.
- **Fixed idle power baseline:** We attribute a constant 7% of the total energy draw to idle Power (GPUs staying on without active workload).

While the tool provides a robust baseline, several limitations remain: the difficulty of tracking exact deployment/undeployment times per model, identifying exactly which GPU hosts which model in a shared environment, and the current lack of data for certain periods (e.g., January 2026).

Strategic Recommendations for DGFIP

In the ‘Tool Methodology’ section, we present in detail an initial version of the tool that allows you to extrapolate energy consumption and GHG emissions results based on inferences across the entire period for which we have usage data. We then outline the limitations of this tool and propose a more refined version that can be used once you have the necessary data. Details of this version are described in the README. The first tool provides an overview of your models, whilst the second will enable a more refined understanding of the environmental impact resulting from the platform’s inference.

Phase IV: End-of-life management

The final stage of the AI lifecycle involves the decommissioning and disposal of hardware. This phase is increasingly critical as the rapid pace of innovation leads to shorter hardware cycles.

There is a significant lack of AI-specific e-waste metrics, making it difficult to track the exact volume and toxicity of hardware discarded by the sector (OECD 2022).

There is also an obsolescence paradox: high-performance GPUs face a shorter amortization window compared to classical servers. While a standard server is typically amortized over 5 to 7 years, the rapid efficiency gains in AI hardware (measured in *gCO₂e/Exaflop*) incentivize replacement after only 3 years (Schwartz et al. 2019).

The dismantling and transport of specialized AI clusters incur a high logistics footprint. Currently, closed-loop recycling, where materials from old GPUs are recovered for new ones—has the potential to offset approximately 4% of the initial manufacturing carbon footprint (Schneider et al. 2025).

Strategic recommendation for DGFIP

- **Lifecycle extension:** Aim to extend the functional lifespan of current GPUs by at least one additional year. This directly reduces the annual impact of the manufacturing debt.
- **Collaborate with your hardware providers,** Dell, to establish and follow recycling methodologies.
- **Quantitative monitoring:** Implement the **NumEcoEval** tool, developed by the CGDD (Commissariat général au développement durable), to precisely quantify the environmental footprint of every hardware component and digital service within the organization **Link to NumEcoEval gitlab.**

Tool methodology

7.1 Overview and data sources

The energy attribution tool estimates electricity consumption and CO₂ emissions per large language model (LLM) deployed by DGFIP. The tool combines two categories of input data:

- **Usage logs** (LiteLLM API gateway): daily per-model aggregates of request counts and token volumes, exported as CSV files. Two files are used: `diane_usage_daily_with_models_2026-01-28.csv` (Oct–Dec 2025, 163 rows, 8 models) and `diane_usage_daily_with_models_fevrier.csv` (Feb 2026, 254 rows, 9 models).
- **Power distribution unit (PDU) data**: electricity measurements from the GPU rack instrumentation. For December 2025, an 80-minute snapshot (`Energy_Data_Clean.csv`, 89 rows, 6 GPU nodes) provides per-node active power in Watts. For February 2026, the file `releves_AI_FEB_2026.csv` (28 991 rows at 5-minute intervals) provides cumulative energy counters in Wh across 6 confirmed circuits. Cumulative data indicates shocks, which result in negative energy per interval. This led us to believe that intervals are actually 10 minute intervals with two energy pathways per voie.

Model architecture parameters (total parameter counts, quantisation format, Mixture-of-Experts configuration) are retrieved from HuggingFace Hub via the safetensors metadata API, with hard-coded fallback values for gated repositories.

7.2 Energy measurement

February 2026 — direct measurement

For February 2026, total IT energy is derived directly from the cumulative PDU counter via consecutive differences. For each circuit c and timestamp t :

Cumulative counter method

$$\Delta W_{c,t} = \text{counter}(c, t) - \text{counter}(c, t - 1)$$

$$E_{\text{IT}} = \frac{1}{1000} \sum_c \sum_{\Delta W_{c,t} \geq 0} \Delta W_{c,t} \quad [\text{kWh}]$$

Reset events ($\Delta W < 0$, corresponding to counter overflow) are discarded.

February 2026 result: 253 resets filtered; $E_{\text{IT}} = \mathbf{5,165}$ kWh (6 circuits confirmed).

December 2025 — power $\times$ uptime method

The December PDU snapshot covers only 80 minutes (3 December 2025). Cluster uptime between observation dates is unknown, so three uptime scenarios are used:

December energy scenarios

$$E_{\text{IT,Dec}} = P_{\text{cluster}} \times 24 \times n_{\text{days}} / 1000 \quad [\text{kWh}]$$

where $P_{\text{cluster}} = 8,778 \text{ W}$ (sum of per-node means across 6 A100 nodes).

Scenario	n_{days}	$E_{\text{IT,Dec}}$ (kWh)
Optimistic (off when idle)	54	11 376
Central (average)	72	15 274
Pessimistic (24/7)	91	19 171

Facility energy and CO₂

Facility energy accounts for datacenter overhead (cooling, power conversion, networking) via the Power Usage Effectiveness (PUE) ratio:

Facility energy and carbon footprint

$$E_{\text{facility}} = E_{\text{IT}} \times \underbrace{1.7}_{\text{PUE}} \quad \text{CO}_2 = E_{\text{facility}} \times \underbrace{0.052}_{\text{kg CO}_2\text{e/kWh}}$$

The CO₂ intensity factor of 0.052 kg CO₂e/kWh is the French national grid annual average for 2024 (RTE eco2mix).

7.3 Per-model attribution

Total cluster energy cannot be measured per model directly. The tool attributes facility energy to each model proportionally to a **computational weight** w_i that proxies the FLOPs contributed by each model. Three corrections are applied relative to a naive tokens-only attribution.

Correction 1 — Numerical precision

Quantised models perform fewer floating-point operations per token. The quantisation factor q_i is defined relative to FP16:

Correction 1: quantisation factor q_i

$$w_i^{(1)} = \text{tokens}_i \times \text{params}_i \times q_i$$

Format	q_i	Derivation	Models
BF16 / FP16	1.000	Reference	Llama-3.3-70B, qwen3vl, Whisper, gte
FP8	0.500	8/16	Qwen-Coder-32B, Mistral-Small-24B
MXFP4-MoE	0.339	$0.10 \times 1 + 0.90 \times (4.25/16)$	gptoss120b, gptoss20b

The MXFP4 factor accounts for 4-bit mantissa with shared floating-point exponent (effective 4.25 bits) applied to 90% of weights, with the remaining 10% retained in FP16.

Correction 2 — Mixture-of-Experts active parameters

Mixture-of-Experts (MoE) models contain N specialised sub-networks (experts) but activate only the top- K per token during inference. The effective parameter count per token is therefore much smaller than the total:

Correction 2: MoE effective parameters

$$\text{eff_params}_i = (1 - f_{\text{MoE}}) \cdot \text{params}_i + \frac{K}{N} \cdot f_{\text{MoE}} \cdot \text{params}_i$$

where $f_{\text{MoE}} = 0.90$ (fraction of parameters in MoE layers).

Model	N experts	Top- K	Total params	Effective params
gptoss120b	128	4	120.4 B	15.4 B (−87%)
gptoss20b	32	4	21.5 B	4.6 B (−79%)

Correction 3 — GPU idle power

NVIDIA A100 GPUs are locked in state P0 (maximum clock frequency) while a CUDA context is active and cannot transition to lower-power states. This results in a persistent idle draw of approximately 50 W per GPU regardless of inference load (Cornelius et al., SC25 2025; confirmed at 62 W by nvidia-smi at 0% utilisation). This idle consumption must be allocated by *hosting time*, not by token volume.

Correction 3: idle power allocation

$$s_i = \underbrace{(1 - \gamma)}_{93\%} \cdot \frac{w_i}{\sum_j w_j} + \underbrace{\gamma}_{7\%} \cdot \frac{d_i}{\sum_j d_j}$$

where $\gamma = 0.07$ (idle fraction, conservative relative to 12–21% reported in literature), $w_i = \text{tokens}_i \times \text{eff_params}_i \times q_i$ is the corrected computational weight, and d_i is the number of days model i was loaded in GPU memory.

7.4 Final energy and CO₂ per model

Attribution result

$$E_i = s_i \times E_{\text{facility}} \quad \text{CO}_{2,i} = E_i \times 0.052$$

Efficiency metrics derived from the attribution:

$$\text{kWh/Mtok}_i = \frac{E_i}{\text{tokens}_i / 10^6} \quad \text{Wh/req}_i = \frac{E_i \times 1000}{\text{requests}_i} \quad \text{gCO}_2/\text{req}_i = \text{Wh/req}_i \times 0.052$$

7.5 February 2026 results

Model	Share	kWh	CO ₂ (kg)	kWh/Mtok	Wh/req	gCO ₂ /req
Llama-3.3-70B-128k	60%	5 233	272	16.91	4.82	0.251
qwen3vl32binstruct	28%	2 447	127	7.91	2.25	0.117
gptoss120b	6%	484	25	1.57	0.45	0.023
gte-Qwen2-1.5B-instruct	3%	240	13	0.78	0.21	0.011
gptoss20b	2%	214	11	0.69	0.20	0.010
deepdml-Whisper-v3-turbo	2%	160	8	0.52	0.15	0.008
Total	100%	8 780	457	4.73	1.34	0.070

Table 3: Per-model energy and CO₂ attribution, February 2026 (1–24 Feb). Facility energy, PUE = 1.7, CO₂ = 0.052 kg/kWh. Three models had zero February usage (Mistral, Qwen-Coder, dgfiP-e5).

7.6 Limitations and known uncertainties

- **No input/output token split** : output tokens cost approximately 2× more FLOPs than input tokens. Without this split, attribution is systematically biased for models with different generation ratios.
- **December snapshot limited to 80 minutes** the December power reference ($P_{\text{cluster}} = 8,778 \text{ W}$) may not be representative of the full Oct–Dec workload, driving a $\pm 34\%$ range in December-period estimates across uptime scenarios.

Strategic recommendation for DGFIP

To transition the current energy monitoring framework from a proof-of-concept to a production-grade system, we recommend the following technical and operational enhancements:

- **Transition to GPU-Level monitoring:** Replace rack-level PDU energy data with granular per-GPU power draw using the `nvidia-smi` interface. This is essential for disaggregating energy consumption when multiple models share the same infrastructure.
- **Enhance data granularity:** Update the usage export pipeline to provide model-level token and request data at 5-minute intervals. This will allow the tool to accurately identify server idle periods and subtract “idle energy”, which currently overstates active emissions by approximately 10%.
- **Differentiate token impact:** Modify the assessment tool to distinguish between **input (prompt)** and **output (generation)** tokens. Since output tokens are significantly more computationally intensive, this split is required for a more precise carbon footprint per request.
- **Automate environmental context:** Integrate the tool with historical RTE carbon intensity archives to ensure accurate reporting for periods older than 6 weeks. For future deployments, maintain a synchronized mapping between internal models and Hugging Face metadata to automate parameter and quantization lookups. => Refer to the README file for details of the refined tool.

Conclusion

8.1 Conclusion on all the LCA

This assessment has provided a holistic view of the environmental impact of AI activities within the DGFIP. By moving beyond simple energy metrics and adopting a multi-criteria Life-Cycle Assessment (LCA), we have identified leverage points to align the development of AI at DGFIP with sustainability goals.

Our analysis of the AI lifecycle highlights four primary areas for intervention:

Hardware and infrastructure: Significant impacts are embedded in the manufacturing phase. Extending the lifespan of your GPUs and improving data center efficiency through optimized PUE and advanced cooling remain the most effective strategies to reduce this burden.

Initial model training: Future procurement should prioritize models with transparent environmental reporting, specifically looking at indicators such as model size, training location (grid carbon intensity), and hardware efficiency.

Inference phase: As the most energy-intensive phase over time, you need to deploy GPU-level monitoring and use our developed assessment tools to refine emission estimates and compare model efficiency in real-time.

End-of-life: Integrating recycling strategies and planning for long-term hardware amortisation are essential to mitigate the obsolescence paradox driven by rapid technological gains.

8.2 Final recommendations and extensions

To support future decision-making, we recommend utilizing the following open-source initiatives and tools:

- **Ecologits Carbon Tracker:** An open-source tool designed to estimate the energy consumption and environmental footprint of generative AI models. **Ecologits Calculator**
- **AI Energy Score:** An initiative providing comparable energy efficiency ratings for AI models. Since they utilize the same H100 GPUs as DGFIP, their ratings are ideal for benchmarking our internal model types. **AI Energy Score Leaderboard**

All materials associated with this project have been centralized in a dedicated GitHub repository.

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